

Seungjae Kim

M.S. Student in Software Convergence

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Research Interests

Video Understanding

- Multi-Object Tracking, Motion Prediction

2D/3D Perception

- Cross-Domain Few-Shot Object Detection, 3D Object Detection

Novel View Synthesis

- 3D Gaussian Splatting, Multi-View Representation Learning

Education

Kyung Hee University

M.S. IN SOFTWARE CONVERGENCE

- Visual Science Lab
- Advisor: Prof. Myeong-Ah Cho

Gyeonggi, South Korea

Sep. 2025 – Aug. 2027 (Expected)

Hanshin University

B.S. IN INFORMATION AND COMMUNICATION ENGINEERING

- Completed mandatory military service (2020 – 2021)

Gyeonggi, South Korea

Mar. 2019 – Aug. 2025

Research Experience

Visual Science Lab, Kyung Hee University

UNDERGRADUATE INTERN

- Research topics: Multi-Object Tracking, Few-Shot Object Detection, and 3D Object Detection
- Advisor: Prof. Myeong-Ah Cho

Gyeonggi, South Korea

Jul. 2024 – Aug. 2025

Publications

International Conference

[1] Rethinking Prototype-based Similarity Learning for Few-Shot Object Detection

KunHo Heo*, **Seungjae Kim***, Wonyu Lee*, SuYeon Kim, MyeongAh Cho *Equal contribution
Under Review

[2] DB-Fusion: Dual-Balanced Sensor Fusion for Multimodal 3D Object Detection

SeungJoon Lee*, Taeyoung Kim*, Useok Choi*, **Seungjae Kim**, Chanse Oh, MyeongAh Cho *Equal contribution
Under Review

[3] PlugTrack: Multi-Perceptive Motion Analysis for Adaptive Fusion in Multi-Object Tracking

Seungjae Kim, SeungJoon Lee, MyeongAh Cho
Proceedings of the **AAAI** Conference on Artificial Intelligence (AAAI), 2026

Domestic Conference

[1] Efficient 3D Gaussian Splatting with Adaptive Depth Refinement in High-Confidence Regions

Seungjae Kim, Taeyoung Kim, YongHyun Gwon, MyeongAh Cho
Korea Software Congress (KSC), 2025

[2] Explainable Intelligent Surveillance Video System Using Spatial Localization

MyoungChul Kim, **Seungjae Kim**, SuYeon Kim, MyeongAh Cho
Korea Software Congress (KSC), 2025

[3] BEV Segmentation Prior-Guided Class-Wise Attention for 3D Object Detection in Autonomous Driving

SeungJoon Lee, Taeyoung Kim, **Seungjae Kim**, MyeongAh Cho
Korea Software Congress (KSC), 2025

[4]

Future-Aware Nonlinear Prediction Module for Real-Time Multi-Object Tracking
Seungjae Kim, Useok Choi, Hayoung Lee, MyeongAh Cho
Korea Computer Congress (KCC), 2025

[5]

Improving Latent Space in Consistency Distillation via Self-Generated Targets
Useok Choi, Seungjae Kim, Chanse Oh, MyeongAh Cho
Korea Computer Congress (KCC), 2025

[6]

Analysis of Key Conditions for Generalizable Generated Video Detection
Junhee Lee, Seungjae Kim, Seungjae Choi, Wongyu Lee, MyeongAh Cho
Korea Software Congress (KSC), 2024

Patents

Domestic Patent

[1]

Method and Apparatus for Generating 3D Scene Graph
KR Registration No. 10-2025-0188361, South Korea, Dec. 2025

Software Copyright

[1]

Improved Sound Source Localization Model through Audio Signal Enhancement
C-2025-035472, South Korea, 2025

Projects

AI Technology for Recognizing Abnormal Situations in Night and Bad Weather
DEEP LEARNING RESEARCHER
• Image restoration and Data pre-processing under adverse visual conditions

Korea Evaluation Institute of Industrial Technology (KEIT)
Jul. 2024 – Present

Anomaly Pre-Detection for Intelligent Surveillance via Vision-Language Fusion
DEEP LEARNING RESEARCHER
• Vision-language model integration for surveillance anomaly interpretation and response

National Research Foundation of Korea (NRF)
Sep. 2024 – Present

Development of an AGI Platform for a Multisensory Social Robot
DEEP LEARNING RESEARCHER
• Few-shot object detection and cross-domain adaptation for robotic perception

Institute for Information & Communications Technology Promotion (IITP)
Apr. 2025 – Jan. 2026

Multimodal Alignment-Based 3D Perception Foundation Model for Autonomous Driving
DEEP LEARNING RESEARCHER
• 3D object detection and LiDAR-RGB sensor fusion

National IT Industry Promotion Agency (NIPA)
Sep. 2025 – Dec. 2025

Reference

Prof. Myeong-Ah Cho
M.S. ADVISOR
• Assistant Professor, Visual Science Lab
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Kyung Hee University